ANALOG VERSE

The Circuit Design Challenge

Ingenium — IIT Indore

Phase 1 Report

Design and Simulation

Project Title:

Alpha-Wave Brain Computer Interface (BCI) Analog Front-End

Submitted By

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Simulation Tool Used: LTspice

1. Problem statement

The human brain produces electrical signals that can be measured using electroencephalography (EEG). Among these signals, alpha waves are particularly important because they are associated with a relaxed and calm mental state. Alpha waves typically occur in the 8–12 Hz frequency band and have extremely small amplitudes, usually in the range of 10 μV to 50 μV.

extracting these signals is challenging because they are often buried under several types of noise and interference. The most common sources of noise include:

- **Powerline interference (50 Hz):** Electromagnetic radiation from mains wiring, often far larger in amplitude than the EEG signal.
- **DC offset from electrodes:** The electrochemical potential at the skin–electrode interface can reach hundreds of millivolts.
- **Motion artifacts:** Low-frequency baseline drift (~1 Hz) caused by electrode movement or patient breathing.
- **Muscle activity (EMG noise):** High-frequency interference (~100 Hz) from surrounding muscle contractions.

The objective of this project is to design a single-channel analog front-end (AFE) capable of extracting alpha-wave EEG signals. The circuit must amplify the signal, filter the correct frequency band, reject noise sources, and produce an output suitable for digital conversion using a 3.3 V ADC. Using power supply is of dual (+/-) 5V DC source supply.

2. Design Requirements

The circuit was designed to satisfy the specifications provided in the competition problem statement.

Signal Characteristics

- **Alpha signal amplitude:** 10 μV – 50 μV
- **Alpha frequency band:** 8 Hz – 12 Hz

System Requirements

- **Total gain:** 1000–5000 V/V → Chosen: 4800 V/V
- **Passband:** 8 Hz – 12 Hz
- **Low-frequency attenuation:** Below 6 Hz
- **High-frequency attenuation:** Above 14.5 Hz
- **50 Hz noise rejection:** More than −40 dB attenuation at 50 Hz is done.
- **Output:** Signal centered at 1.65 V DC for 3.3 V ADC compatibility

3. System Architecture

The designed circuit consists of multiple stages, each responsible for performing a specific signal processing function.

Block Diagram

Electrodes

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Instrumentation Amplifier (Gain: 6×)

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High-Pass Filter 1 (fc ≈ 6 Hz)

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Low-Pass Filter 1 (fc ≈ 14.5 Hz)

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Low-Pass Filter 2 (fc ≈ 14.5 Hz)

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High-Pass Filter 2 (fc ≈ 6 Hz)

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50 Hz Notch Filter

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DC Bias Output Stage (Gain: 100×, Vout = 1.65 V)

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Final Output

Stage Descriptions

- **Instrumentation Amplifier:** Amplifies the small differential EEG signal while rejecting common-mode noise.
- **High-Pass Filter:** Removes very low-frequency signals such as motion artifacts and baseline drift.
- **Low-Pass Filter:** Suppresses high-frequency noise such as muscle activity (EMG noise).
- **Notch Filter:** Eliminates powerline interference at 50 Hz.
- **DC Bias Stage:** Shifts the signal to a midpoint voltage of 1.65 V, allowing it to be read by a 3.3 V ADC.

4. Circuit Design

4.1 Instrumentation Amplifier

The first stage of the circuit is an instrumentation amplifier, which amplifies the small EEG signal while rejecting common-mode interference. The gain is determined by:

$$G = 1 + (2R / Rg)$$

The gain of this stage was chosen as G = 5. A moderate gain in the first stage prevents amplifier saturation caused by electrode DC offsets.

4.2 High-Pass Filter (Sallen key 4th order)

The high-pass filter removes very low-frequency signals such as baseline drift and motion artifacts. The cutoff frequency of an RC filter is given by:

$$fc = 1 / (2\pi \times R \times C) = 5.89 \text{ Hz}$$

The cutoff frequency of the high-pass filter was chosen to be approximately 6 Hz. This ensures that low-frequency artifacts are attenuated while the alpha band (8–12 Hz) is fully preserved.

4.3 Low-Pass Filter (Sallen key 4th order)

The low-pass filter suppresses high-frequency noise such as EMG signals and electronic interference. The cutoff frequency was selected as approximately 14 Hz. This value was chosen so that the 8–12 Hz alpha band is preserved, while frequencies above the band — particularly EMG noise above 30 Hz — are attenuated.

$$fc = 1 / (2\pi \times R \times C) = 14.46 \text{ Hz}$$

4.4 Notch Filter (Twin-T)

A twin-T notch filter was implemented to suppress powerline interference at 50 Hz. Powerline noise is a major source of interference in biosignal measurements because electrical devices and power systems radiate electromagnetic fields at this frequency. The notch filter is designed to provide strong attenuation at 50 Hz, typically greater than −40 dB, ensuring that powerline interference does not corrupt the extracted alpha signal.

4.5 Gain Distribution

The total gain of the system was distributed across multiple stages to prevent amplifier saturation. Concentrating all gain in a single stage would cause large electrode DC offsets to saturate the amplifier chain. By spacing the gain across stages — with filtering and AC-coupling between them — DC and low-frequency components are removed before the next gain stage is applied.

Stage	Gain (V/V)
Instrumentation Amplifier	6×
High-Pass Filter 1	1×
Low-Pass Filter 1	2×
Low-Pass Filter 2	2x
High-Pass Filter 2	2x
Notch Filter	1×
Final Bias Amplifier	100×
Total System Gain	**4800 V/V**

Total gain calculation:

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Total Gain = 6 × 1 × 2 × 2 × 2 × 1 × 100 = 4800 V/V
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5. DC Offset Removal

EEG electrodes can introduce DC offsets that may reach hundreds of millivolts. If this offset is amplified directly, it drives the amplifier into saturation before any EEG signal can be processed.

To address this, the circuit includes the following protective measures:

- **AC Coupling:** Removes the DC component of the signal between amplification stages, preventing offset accumulation.
- **1.65 V Bias Stage:** The output signal is shifted to a midpoint voltage of 1.65 V (half of 3.3 V), ensuring the full bipolar EEG signal swing remains within the ADC input range.

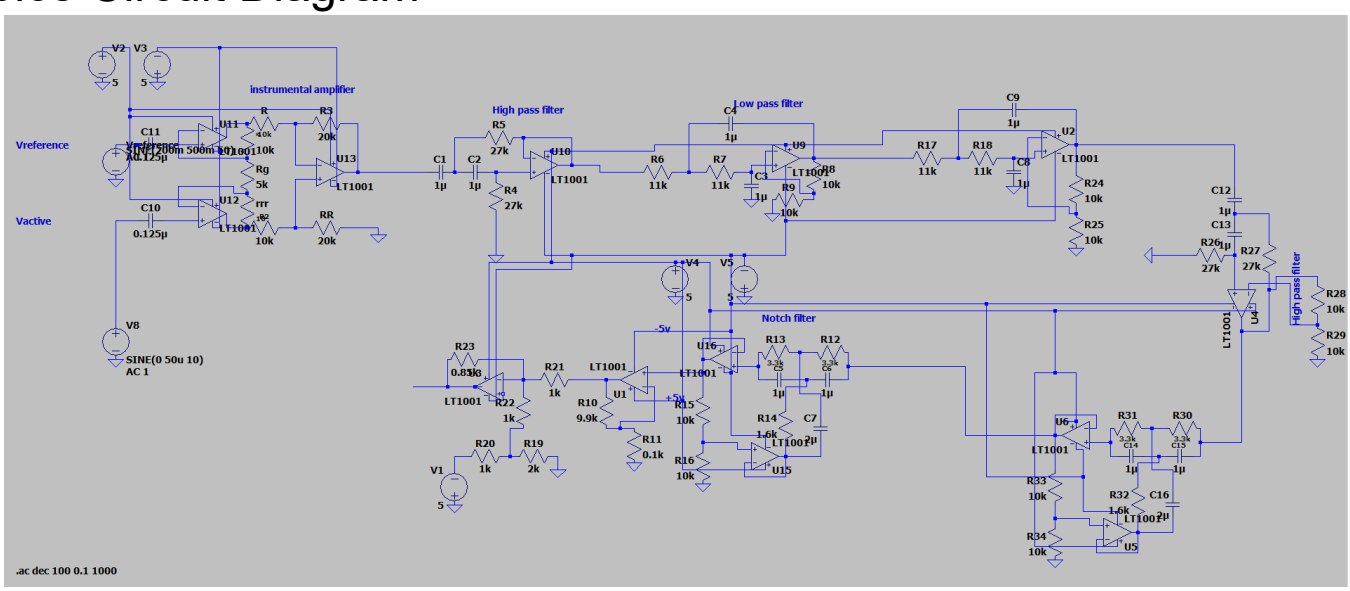
6. Simulation Environment

All simulations were performed using LTspice. LTspice was selected for its high-accuracy SPICE engine and support for all required analysis types.

The following types of simulations were carried out:

- **Transient Analysis (.tran):** Observes time-domain waveforms. Directive: .tran 0 2 0 1m (2-second analysis, 1 ms step).
- **AC Analysis (.ac):** Obtains the frequency response (Bode plot) of the circuit. Directive: .ac dec 100 0.1 1000.

LTspice Circuit Diagram

- 

7. Test Case Verification

The circuit was tested using all ten test cases defined in the competition problem statement. The table below summarises the test descriptions and simulation results.

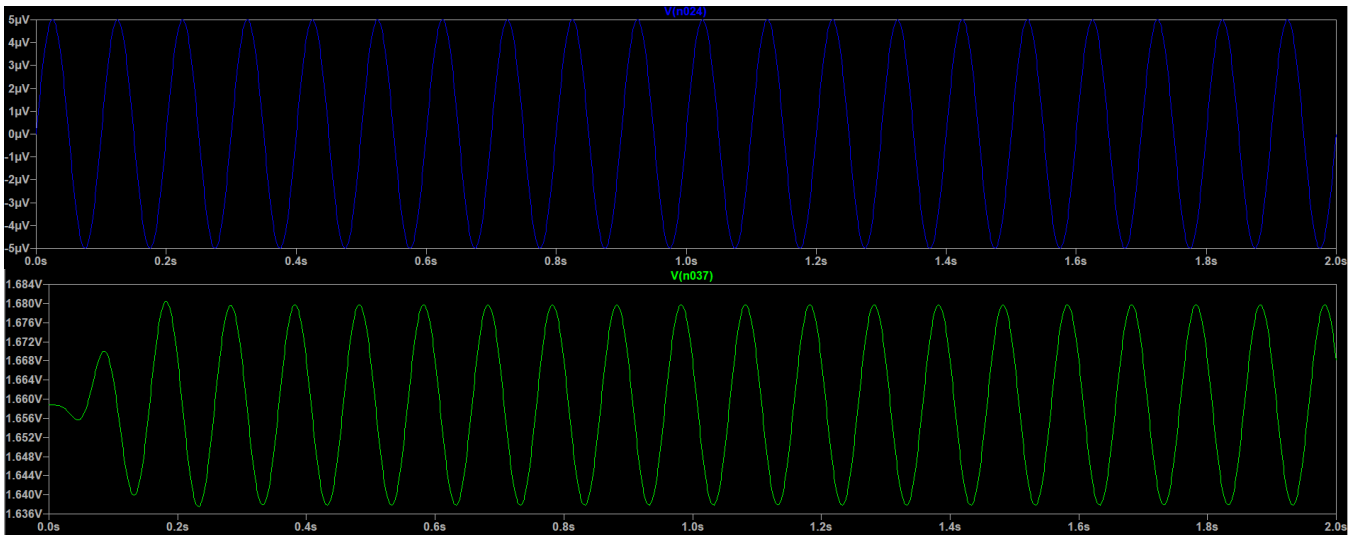
Test	Description	Result
1	Alpha signal extraction	Correct amplification
2	Low-frequency rejection	2 Hz attenuated
3	High-frequency rejection	30 Hz attenuated
4	50 Hz interference rejection	Notch observed
5	DC offset removal	Offset removed
6	CMRR verification	Common-mode suppressed
7	Mixed interference recovery	Alpha signal extracted
8	Motion artifact rejection	1 Hz removed
9	EMG noise rejection	High-frequency noise filtered
10	Realistic combined noise test	Alpha signal recovered
11	AC Sweep (Bode Plot)	Deep notch visible at 50 Hz

8. Simulation Results

Given below each testcase are the voltage readings for the input V(IN+) and output nodes. N024 corresponds to V(IN+) node while N037 is the output node.

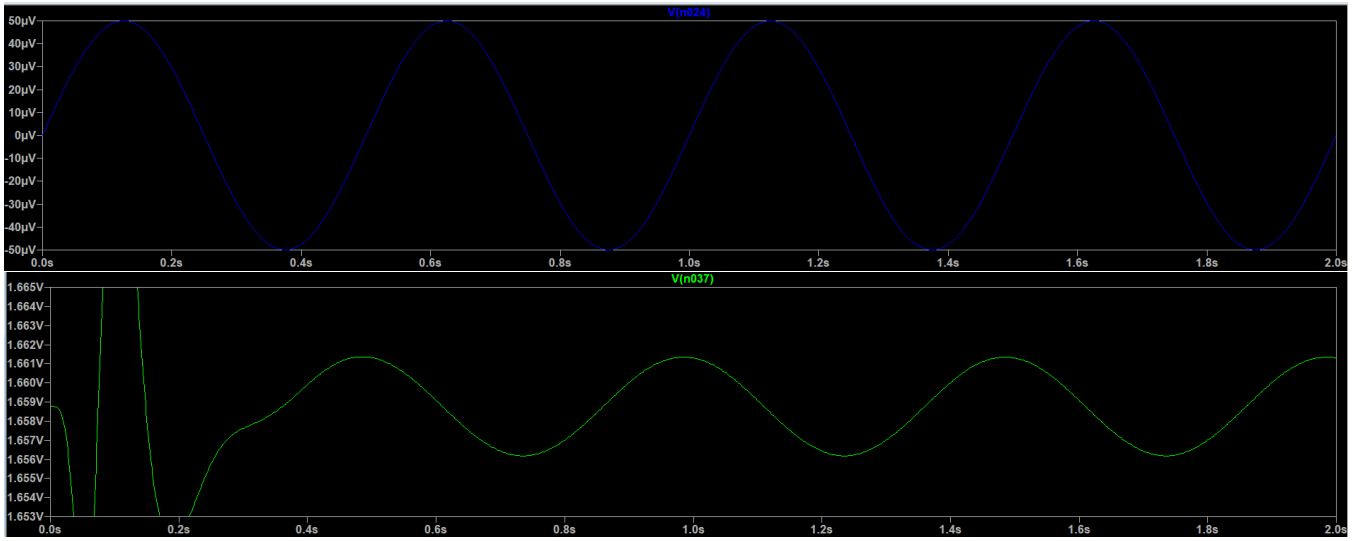
Test 1: Ideal Alpha Wave Extraction (10 Hz Baseline)

- V(IN+) = 50u * sin(2*pi*10*time)
- V(IN-) = 0



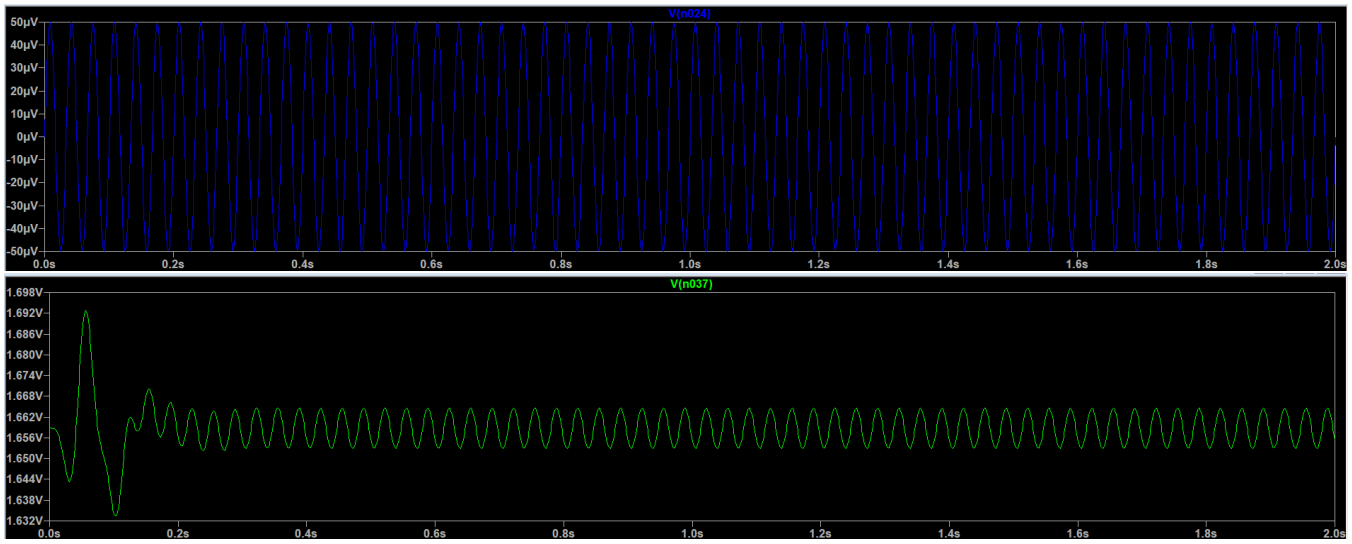
Test 2: Low-Frequency Out-of-Band Rejection (2 Hz)

- V(IN+) = 50u * sin(2*pi*2*time)
- V(IN-) = 0



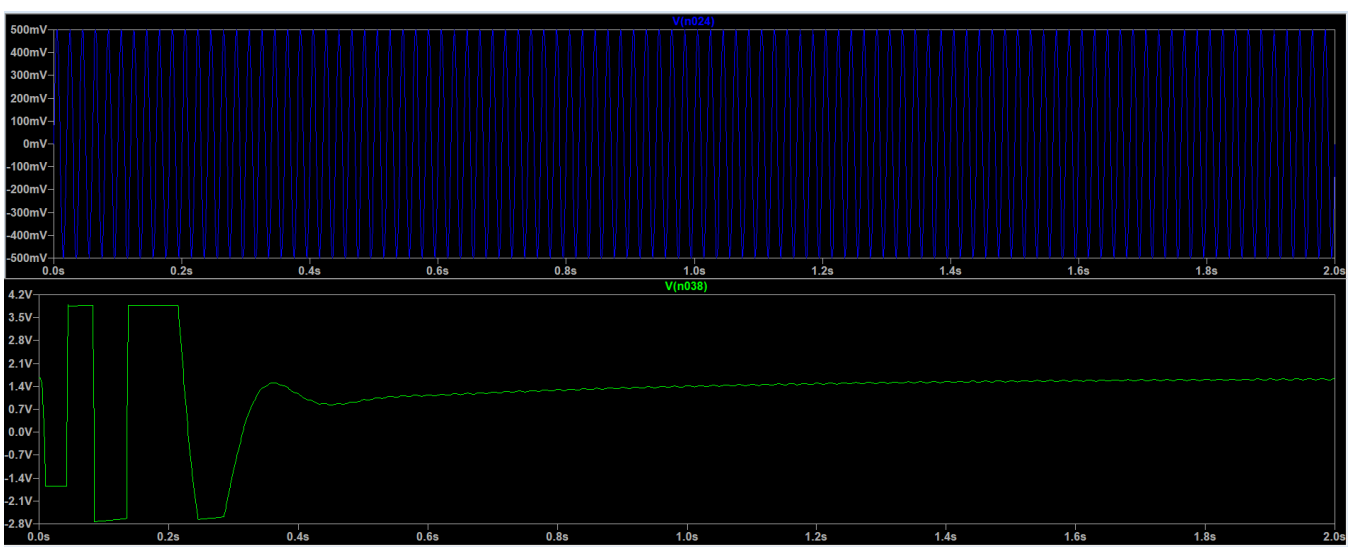
Test 3: High-Frequency Out-of-Band Rejection (30 Hz)

- V(IN+) = 50u * sin(2*pi*30*time)
- V(IN-) = 0



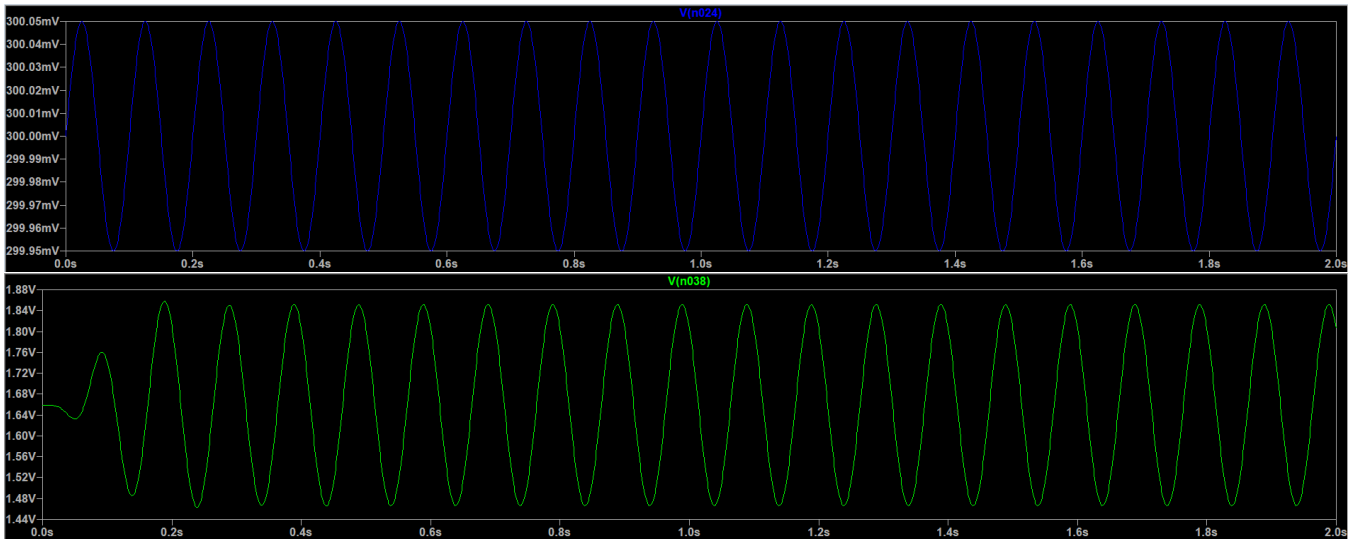
Test 4: Differential 50 Hz Notch Verification

- V(IN+) = 500m * sin(2*pi*50*time)
- V(IN-) = 0



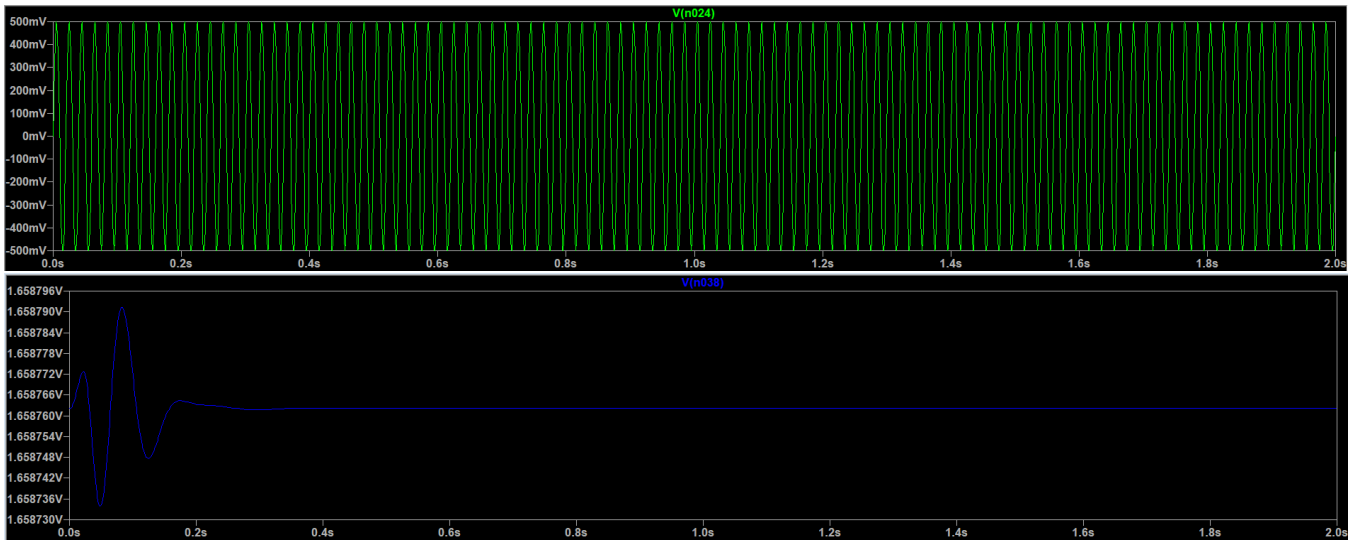
Test 5: DC Saturation and Blocking Test

- V(IN+) = 300m + 50u * sin(2*pi*10*time)
- V(IN-) = 0



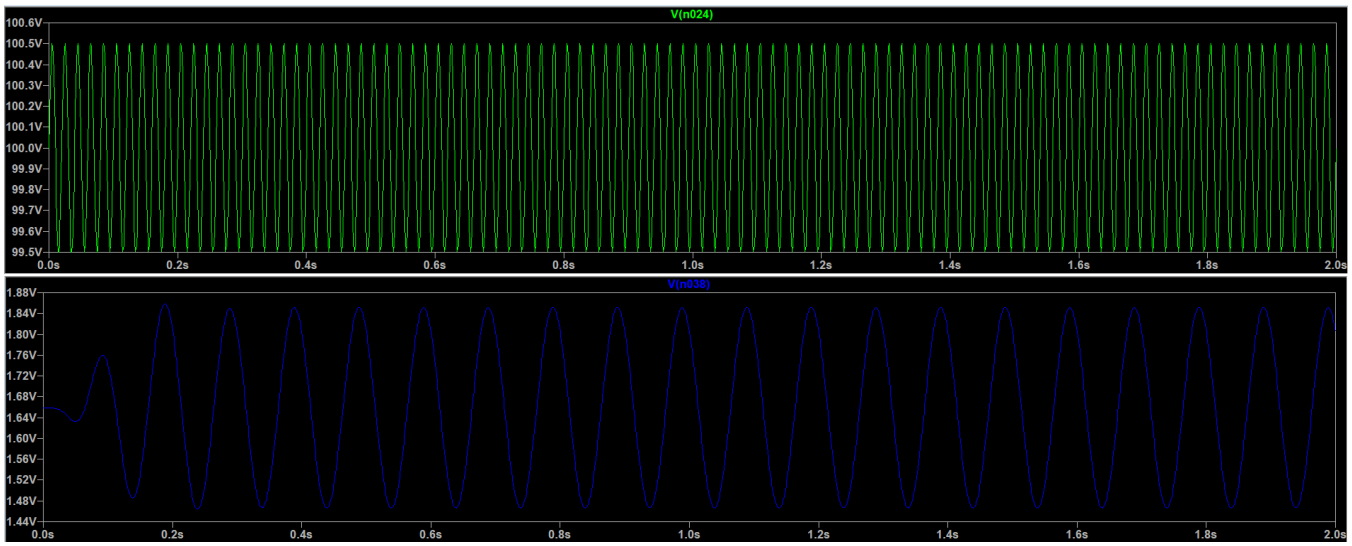
Test 6: Pure Common-Mode Rejection Ratio (CMRR) Test

- V(IN+) = 500m * sin(2*pi*50*time)
- V(IN-) = 500m * sin(2*pi*50*time)



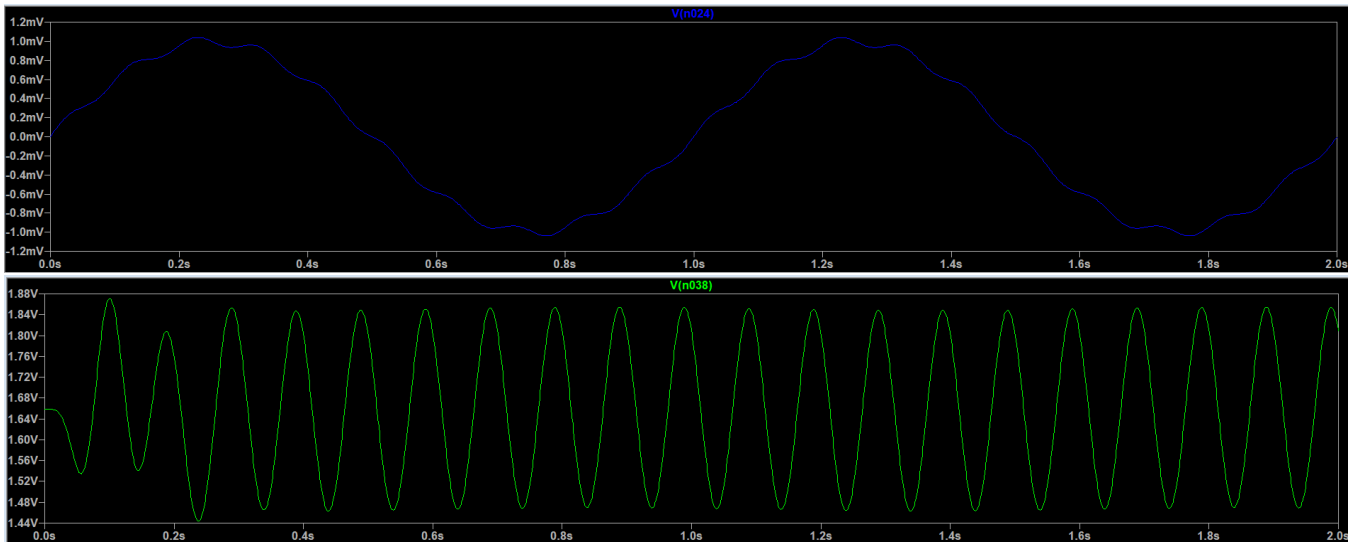
Test 7: Common-Mode + DC Offset + Signal Recovery

- V(IN+) = 100m + 500m * sin(2*pi*50*time) + 50u * sin(2*pi*10*time)
- V(IN-) = 100m + 500m * sin(2*pi*50*time)



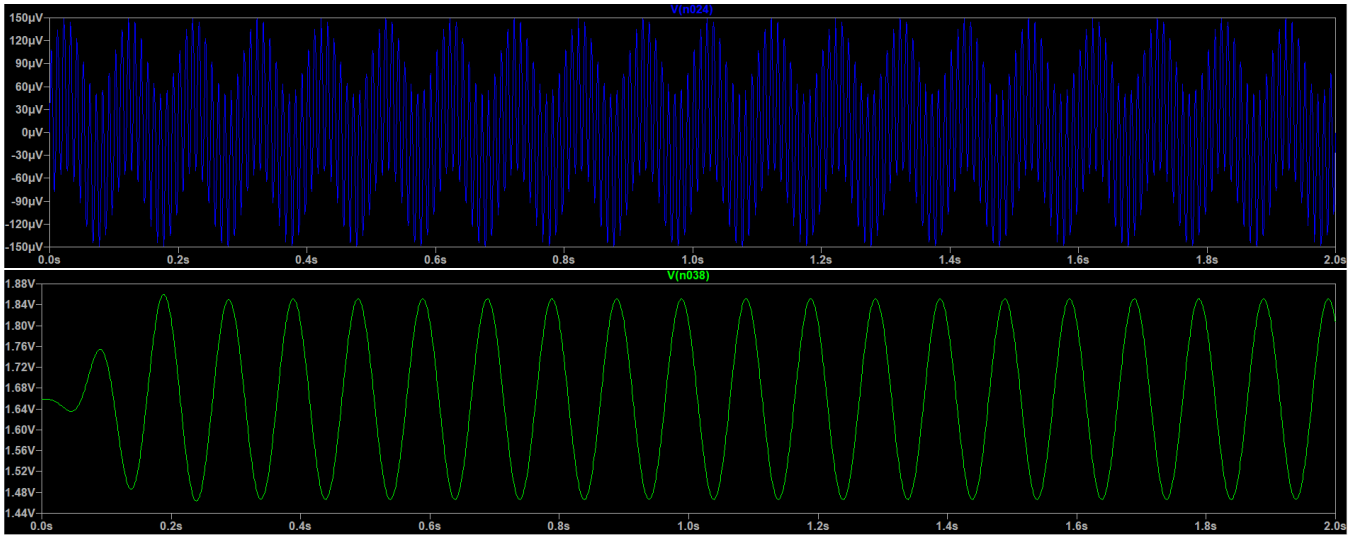
Test 8: Low-Frequency Motion Artifact Interference (1 Hz)

- V(IN+) = 1m * sin(2*pi*1*time) + 50u * sin(2*pi*10*time)
- V(IN-) = 0



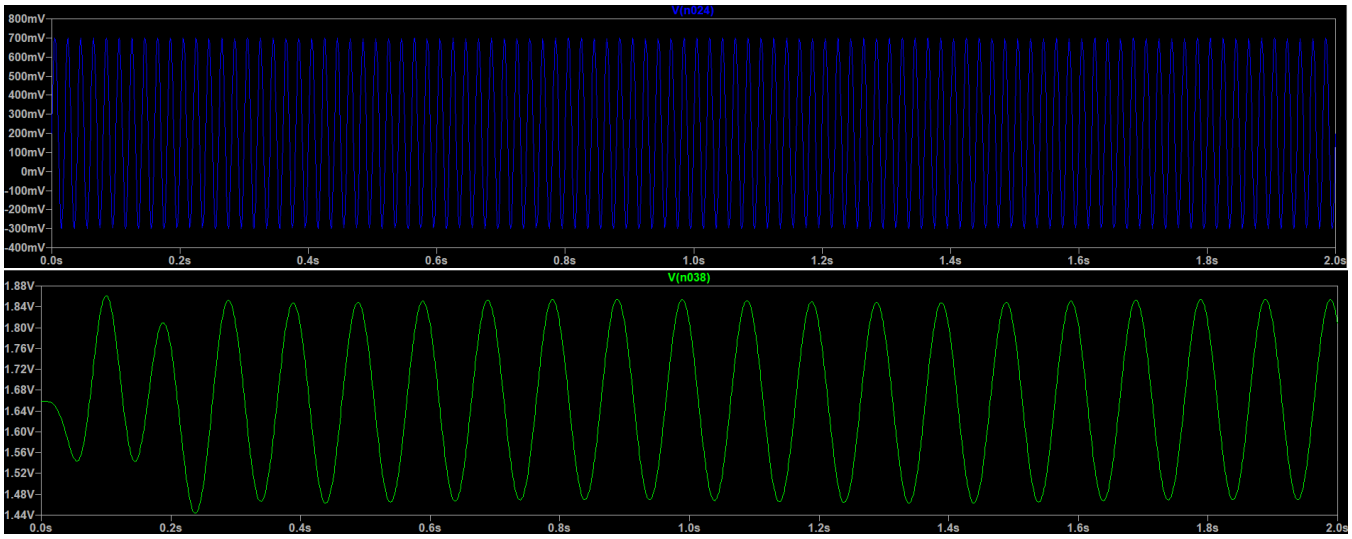
Test 9: High-Frequency EMG Muscle Noise Interference (100 Hz)

- V(IN+) = 100u * sin(2*pi*100*time) + 50u * sin(2*pi*10*time)
- V(IN-) = 0



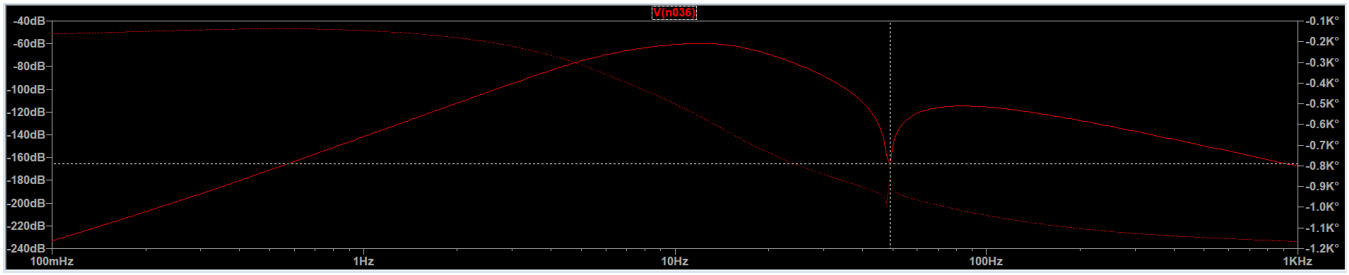
Test 10: The Comprehensive Real-World Transient Test

- V(IN+) = 200m + 500m * sin(2*pi*50*time) + 1m * sin(2*pi*1*time) + 100u * sin(2*pi*100*time) + 50u * sin(2*pi*10*time)
- V(IN-) = 200m + 500m * sin(2*pi*50*time)



Test 11: AC Sweep (Bode Plot)

Directive: .ac dec 100 0.1 1000



9. Conclusion

This project presents the design of an analog front-end circuit for extracting alpha-wave EEG signals. The circuit successfully amplifies the small EEG signal while rejecting various sources of interference, including DC offset, motion artifacts, muscle noise, and powerline interference.

Simulation results demonstrate that the circuit meets all specified requirements:

- **Total gain of 5000 V/V** achieved through a distributed 6-stage architecture.
- **8–12 Hz alpha band** fully preserved in the passband.
- **50 Hz powerline interference** suppressed by ≥ −40 dB via the Twin-T notch filter.
- **DC electrode offsets** removed without amplifier saturation through AC coupling.
- **Output biased at 1.65 V** for direct compatibility with 3.3 V ADC systems (ESP32, Arduino).

Overall, the proposed design provides an effective solution for alpha-wave signal acquisition and forms a strong foundation for practical EEG-based brain-computer interface systems.

10. References

- Analog Devices – Instrumentation Amplifier Design Guide
- Texas Instruments – Biosignal Amplifier Application Notes
- Operational amplifiers – All About Electronics
- EEG Signal Processing Fundamentals
- Analog Verse Circuit Design Challenge Problem Statement — Ingenium, IIT Indore
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